

Combinations

Monday, March 10, 2025 5:57 PM

COMBINATIONS

A **combination** is an arrangement of the items in a set in which the order of the items doesn't matter.

Suppose there are 8 workers in an office. The fire safety team needs 3 people. The manager puts the 8 workers' names on slips of paper and randomly chooses 3 names. Does the order the names are chosen make a difference?

No, the workers will be a group. The order doesn't matter.

This is an example of a combination. Now let's see how to count it.

Leticia	Joshua
Calvin	= Leticia
Joshua	Calvin

The orders are different, but the group of names is the same.

COUNTING COMBINATIONS

Example Three employees out of a group of 8 will be chosen randomly to be the fire safety team. How many different groupings are possible?

Step 1

Find the permutations for choosing 3 names out of 8.

$$8 \times 7 \times 6 = 336$$

There are 336 possible combinations.

Step 2

Find the permutations for a group of 3. You are finding how many different ways 3 names can be arranged.

$$3 \times 2 \times 1 = 6$$

Step 3

Divide.

$$336 \div 6 = 56$$

How can you tell the difference between a permutation and a combination problem? You need to use common sense. Remember, if changing the order matters in the problem, it's a permutation.

No one does this !!
learn & use the
Formulas.

<u>Combination</u>	<u>Permutation</u>
$n C r = \frac{n!}{r!(n-r)!}$	$n P r = \frac{n!}{(n-r)!}$

$$8 C 3 = \frac{8!}{3! 5!} = \frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 1 \times 5!} = 56 \text{ ways}$$

Combinations, cont'd

Monday, March 10, 2025 5:59 PM

Be sure to consider whether order matters.

Decide what kind of problem each situation describes. Write *P* for permutation or *C* for combination.

- | | |
|--|---|
| 1. In a lottery, you win a prize if you can guess which 6 balls will be drawn from 30 balls, numbered 1 through 30. <i>C</i> | 4. Three scoops of different flavors of ice cream are placed in a bowl to make a sundae. <i>C</i> |
| 2. A class president and secretary will be randomly chosen from a class of 20 students. <i>P</i> | 5. A baseball manager chooses 3 players from a group of 15 to bat in the first, second, and third spots in the lineup. <i>P</i> |
| 3. A phone company generates new numbers for an area based on the last four digits in a telephone number. <i>P</i> | 6. Five basketball players are chosen from a group of 12 to start the game. <i>C</i> |

1. Combination. Does not say balls must be ordered.
2. Permutation. It matters who will be president and who will be secretary.
3. Permutation. Order of numbers matters in a person's phone number.
4. Combination. Does not matter the order the flavors are put in the bowl.
5. Permutation. It matters who will be in the 1st spot, then the 2nd spot, and finally, the 3rd spot.
6. Combination. Order of the basketball players does not matter.

Examples

Monday, March 10, 2025 6:00 PM

Solve.

7. From an office with 15 workers, 4 workers are chosen randomly to form a holiday committee. How many different committees could be chosen? Combination

$$nCr = n! / [r! (n-r)!]$$

$$15C4 = 15! / [4! 11!]$$

$$= \frac{(15)(14)(13)(12)(11!)}{(4)(3)(2)(1)(11!)}$$

= 1365 committees

8. Mark has 6 different T-shirts. He puts 2 in a suitcase to take on a trip. How many different groups of 2 T-shirts are possible? combination

$$nCr = n! / [r! (n-r)!]$$

$$6C2 = 6! / [2! 4!]$$

$$= \frac{6 \times 5 \times 4!}{2 \times 4!}$$

= 15 groups

9. A study group of 3 students is formed from a class of 18 students. How many different study groups can be formed? combination

$$nCr = n! / [r! (n-r)!]$$

$$18C3 = 18! / [3! 15!]$$

$$= \frac{18 \times 17 \times 16 \times 15!}{3 \times 2 \times 1 \times 15!}$$

= 816 groups

10. A florist has 6 different kinds of flowers to choose from. If he makes a bouquet using 3 kinds of flowers, how many combinations are possible?

$$nCr = n! / [r! (n-r)!]$$

$$6C3 = 6! / [3! 3!]$$

$$= \frac{(6)(5)(4)(3!)}{(3)(2)(3!)}$$

= 20 bouquets

11. A magician tells an audience member to choose 2 cards from a deck of 52 different cards. How many different 2-card pairings are possible? combination

$$nCr = n! / [r! (n-r)!]$$

$$52C2 = 52! / [2! 50!]$$

$$= \frac{52 \times 51 \times 50!}{2 \times 50!}$$

= 1326 pairings

12. On a reality show, 5 people will be chosen from a group of 30 people to go on to the next round. How many different groups are possible for the next round? combination

$$nCr = n! / [r! (n-r)!]$$

$$30C5 = 30! / [5! 25!]$$

$$= \frac{30 \times 29 \times 28 \times 27 \times 26 \times 25!}{5 \times 4 \times 3 \times 2 \times 1 \times 25!}$$

= 142,506 groups

Solve. The problems may involve permutations or combinations.

13. In a cooking contest, teams of chefs will have the chance to cook for a panel of judges. The teams of 3 will be formed randomly from a pool of 20 chefs.

- a. How many different groups of 3 can be formed?

COMBINATION

$${}_{20}C_3 = \frac{20!}{3!17!}$$

$$= \frac{20 \times 19 \times 18 \times 17!}{3 \times 2 \times 1 \times 17!}$$

$$= 1140 \text{ teams}$$

- b. Five teams will take turns competing the first night of the contest. In how many different ways could 5 teams be scheduled?

PERMUTATION

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}_5P_5 = \frac{5!}{0!} \text{ Remember } 0! = 1$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

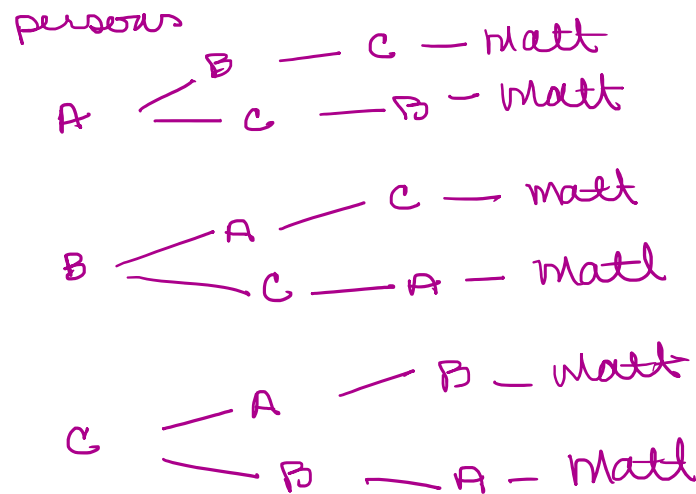
$$= 120 \text{ ways}$$

14. A swim coach chose 4 swimmers for the relay race. He definitely wants Matt to swim the last leg of the race. In how many different orders could the swimmers swim the relay race?

PERMUTATION

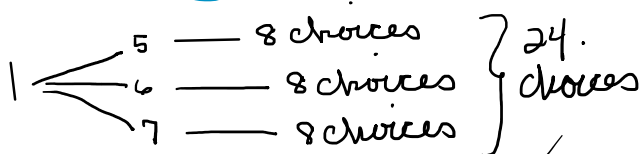
$${}_3P_3 = \frac{3!}{0!}$$

$$= 6 \text{ ways}$$



15. In a small office, every employee is assigned a sign-in code. The code has 3 digits. The first digit cannot be a 0. The second digit must be a 5, 6, or 7. The third digit must be different from the first digits. How many 3-digit codes are possible?

Permutation



1
.
.
.
9

times 9 is

$$\therefore 9 \times 3 \times 8 = 216 \text{ codes}$$

1st digit	2nd digit	3rd digit
1..9	5, 6, or 7	0..9

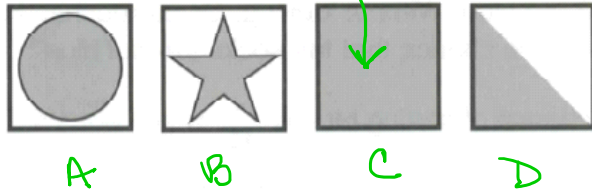
But different

${}_9P_1$	${}_3P_1$	${}_8P_1$
$= \frac{9!}{8!}$	$= \frac{3!}{2!}$	$= \frac{8!}{7!}$
$= 9$	$= 3$	$= 8$

9 times 4 is
216 possible codes

... 4 x 0 x 0 = 216 codes

16. The blocks below are placed in a row. In how many different orders can you place the blocks if the fully shaded block cannot be placed at either end of the row?

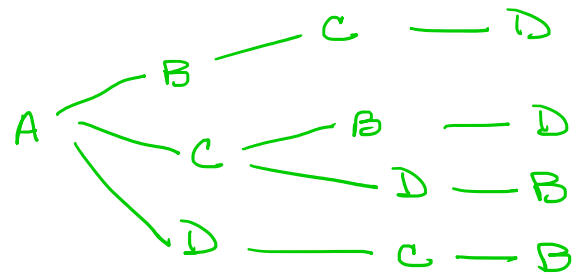


permutation

with C as the 2nd block: $3P_3 = 3!/0! = 6$

with C as the 3rd block: $3P_3 = 3!/0! = 6$

$6 + 6 = 12$ ways



4 ways

&

Repeat for B & D
∴ 12 ways